FTC EVENT SCOUT

Native macOS scouting for FIRST Tech Challenge

01

PROBLEM DEFINITION

FTC scouts need to compare many teams and understand the matches behind the rankings. One rank cannot show autonomous strengths, alliance context or abnormal scores. The project brings event data and observations together to support informed scouting decisions.

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SUCCESS CRITERIA

Observable project targets - the scout can:

- Load teams and scored matches for an event.
- Sort rankings and compare available OPR metrics.
- Open charts, team histories and score breakdowns.
- Show predictions with data-quality labels.
- Save and reload color-coded local scouting tags.



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PROBLEM SCENARIO

At an FTC event, a scout selects a season and event code, compares teams and investigates match evidence.

Languages & apps: Swift / SwiftUI for macOS; a local Python 3 helper; FIRST Events data processed into CSV. FTCScout supplies pre-event history.

Methods & algorithms: least-squares OPR (Offensive Power Rating), with one-sided Jacobi SVD in the Swift prediction layer. OPRc filters alliance residuals once using 1.5 x IQR (interquartile range), then recalculates OPR. Match estimates sum team OPRc values.

04

THE SOLUTION

FTC Event Scout combines sortable rankings, OPR charts, event highlights, team histories and detailed score breakdowns in one native app.

Scouts compare Auto, Teleop, Endgame and overall contribution; inspect the alliances behind a score; and save reusable, color-coded team tags locally.

The Predictions view estimates red and blue alliance scores, winner and signed margin for scheduled or custom four-team matchups.

IMPORTANT CONSIDERATIONS

OPR and OPRc are scouting estimates. Defense, alliance interaction and robot failures can change outcomes. Predictions flag limited data; missing team estimates make a prediction unavailable.

macOS 14+ | Python 3 | FIRST Events API credentials

github.com/Sakzyo/FTC_Event_Scout